

Propeller Ground Effect in Forward Flight

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The study of the fixed-pitch small-scale propeller in-ground effect is extended to forward flight conditions at different incidence angles of the propeller. Force-based experiments, smoke flow visualization, surface flow visualization, and phase-locked particle image velocimetry (PIV) indicate the presence of three distinct performance zones as a function of the advance ratio and ground distance, where the ground effect changes drastically. The power required at constant thrust shows an increasing trend after a threshold advance ratio for each h/D and then decreases steeply upon a further increase in the advance ratio. Phase-locked PIV revealed the presence of a fountain vortex and a ground vortex at distinct J for each h/D distance that correlates with an increase in power and a decrease in thrust, respectively. At advance ratios below the onset of the fountain vortex lies the optimal ground effect zone. An increase in the advance ratio beyond this threshold will result in power increments due to the formation of the fountain vortex. A further increase in the advance ratio initiates the onset of the ground vortex where the ground effect is significantly reduced. The onset of the fountain vortex and the ground vortex occurs at comparatively lower advance ratios for propellers at a positive incidence angle. A normalization method is developed for ground effect forward flight that collapses the in-ground-effect performance for various normalized heights, advance ratios, and incidence angles.

Nomenclature

C_P	=	power coefficient; $2\pi C_Q = (P/\rho n^3 D^5)$
C_Q	=	propeller torque coefficient; $(Q/\rho n^2 D^5)$
C_T	=	propeller thrust coefficient; $(T/\rho n^2 D^4)$
D	=	propeller diameter, m
h	=	distance to the ground plate, m
J	=	advance ratio; V_∞/nD
J_S	=	scaled advance ratio
J^*	=	normalized advance ratio; $J\sqrt{(\pi/2C_T)}$
n	=	propeller rotational speed per second
P	=	power, W
Q	=	torque, N/m
R	=	propeller radius, m
T	=	thrust, N
V	=	flow velocity, m/s
V_i	=	propeller induced velocity, m/s
V_w	=	propeller wake velocity, m/s
V_∞	=	freestream velocity, m/s
γ	=	propeller pitch, m
δ	=	propeller forward tile angle, deg
θ	=	propeller incidence angle, deg
ρ	=	air density, kg/m ³
σ	=	propeller solidity; $S/0.25\pi D^2$
$\chi_{\text{eff,IGE}}$	=	effective propeller wake skew angle, deg
χ_{prop}	=	propeller wake skew angle, deg

Subscripts

IGE	=	in-ground effect
OGE	=	out-of-ground effect
$x,$	=	body frame coordinates
$X,$	=	inertial frame coordinates

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I. Introduction

GROUND effect on lifting rotor/propeller performance during hovering has been studied theoretically [1] and experimentally [2–6] for decades. Placing the propulsor adjacent and parallel to a ground plate offers two benefits: an increase in the thrust produced and a decrease in the power required, for a given rotational speed. However, the latter benefit is usually ignored in many small-scale propeller-driven unmanned aerial vehicle (UAV) studies. Multiple ground effect models for hovering have been developed for large rotors, including Betz's ground effect model [1], Cheeseman and Bennett's model developed [2] based on Betz's theory [1], and Hayden's model developed based on experimental data [4]. However, most of the models do not produce a good prediction on small-scale fixed-pitch propellers widely used by smaller-scale UAVs, especially at a small distance to the ground, which was discussed in prior studies by Cai et al. [7,8].

The rotor ground effect in forward flight has also been widely studied for larger rotors analytically and experimentally with wind-tunnel tests and flight tests [2,5,6,9–16]. Cheeseman and Bennett [2] conducted the flight test on a full-scale Sikorsky S-51 in hover and forward flight in which the in-ground-effect (IGE) power required at constant thrust P_{IGE} was normalized by the power required at the out-of-ground effect (OGE) hovering condition ($J = 0$) to represent the performance of the rotor in-ground effect. An increment in $P_{\text{IGE}}/P_{\text{OGE}}$, with the increment of J at $J < 0.1$ found in $h/D \leq 0.3$, resulted in a higher IGE power consumption for the helicopter at low-speed forward flight due to the loss in rotor performance. Sheridan and Wiesner [5] conducted wind-tunnel experiments on a scaled model of the Boeing CH-46 Vertol helicopter. Flow visualization was conducted to investigate the effect of the ground vortex between the rotor and the ground plate on the higher rotor power consumption and the changes in the trim condition. A rotor leading-edge recirculation/vortical flow region was observed at low advance ratios, where the rotor power consumption increased with constant thrust.

A similar IGE flow pattern is also observed in the study conducted by Curtiss et al. [6] on a 1.22 m four-blade rotor. The normalized advance ratio J^* is also introduced in the study. J^* takes the collective pitch of the blade and the resultant change in C_T into account when normalizing the freestream velocity. J^* also represents the ratio of the freestream velocity V_∞ and the propeller-induced velocity V_i , which determines the gross characteristics of the rotor wake. At $J^* < \sim 0.45$, the flow pattern between the rotor and ground is similar to the IGE hovering condition. Within the range of $0.45 < J^* < 0.7$, the aforementioned rotor leading-edge recirculation is observed. With the further increment of the freestream velocity, within the range of $0.7 < J^* < 1.0$, the ground vortex is observed. Similar results are

also found in the studies conducted by Schwarz et al. [10] and Nathan and Green [11]. However, Curtiss et al. [6] did not specify the upper bound of J^* where the ground vortex vanishes.

Schwarz et al. [10] performed particle image velocimetry (PIV) on a Mikado Logo 30 SP model helicopter with a 1.4 m rotor diameter to investigate the IGE ground vortex strength. At $J^* \sim 0.7$, where the transition between the fountain vortex and ground vortex occurs, the strength of the fountain/ground vortex reaches its maximum, resulting in a higher unsteadiness of the rotor. Based on the PIV results from Ganesh and Komerath [13], the maximum strength of the ground vortex could be up to four times the strength of the blade tip vortex.

Nathan and Green [11] conducted wind-tunnel experiments with PIV on a 1.0 m four-blade rotor with a moving belt, which better simulates the actual flight condition. The location of the ground vortex was found to be further downstream with the moving belt when compared to the traditional stationary ground plate. A higher J^* was also tested to determine the upper bound of J^* where the ground vortex vanishes. For a stationary ground plate, the ground vortex vanishes at $J^* \sim 1.2$, whereas the ground vortex vanishes at a higher J^* of 1.6 when using the moving belt. Moreover, based on the study conducted by Rae [17], the area ratio between the wind tunnel and the testing model also affects the location of the ground vortex when testing in a closed-jet configuration.

Almost all the literature discussed earlier in this paper used a conventional rotor with a relatively small spanwise twist or taper. However, in prior work [7,8], the performance of a propeller with a twist and taper distribution was found to be different from that of the larger rotors, especially in-ground effect. The IGE performance difference between a conventional rotor and a propeller blade arises from the unique geometry of the fixed-pitch propeller. The pitch-to-diameter ratio was found to have a large impact on the propeller performance in-ground effect in hover [8]. Modern UAVs and drone designs use a propeller rather than a conventional rotor geometry as a main propulsive device. Given the drastic changes between the propeller and the rotor in IGE hover performance [7,8], as well as the overall difference in the OGE edgewise forward flight performance [18], it is essential to extend the investigation to study the propeller IGE performance in forward flight. The current investigation extends the prior study to include the propeller ground effect performance in forward flight at two different incidence angles using force-based, flow visualization, and optical flow diagnostic experiments. By comparing the propeller results with the standard rotor results from the literature, the role of propeller geometry in the ground effect forward flight performance will be elucidated.

II. Experimental Setup

A. Force-Based Experiment

Experiments were conducted in the University of Dayton Low-Speed Wind Tunnel (UD-LSWT) in the open jet configuration. The UD-LSWT has a 16:1 contraction ratio, six antiturbulence screens, and 0.762 by 0.762 m open jet test sections. A 0.914 by 1.829 m aluminum alloy sheet with 0.005 m thickness was used to simulate the ground plate for the experiment. All propellers were driven by an E-Flite Power 60–400 Kv brushless outrunner electric motor controlled by a Hobbywing Platinum V4 Electronic Speed Controllers (ESC) under governor mode to maintain constant rotation speed. The propeller and motor were driven by a PSW 30–108 constant-voltage power supply. A photodiode was used to measure propeller revolutions per minute (RPMs). The schematic of the setup is given in Fig. 1. The variation of the propeller rotational speed within the same experimental run is less than 1%, which was verified by the data collected by the photodiode.

The propeller motor was attached to an ATI Industrial Automation Mini-40 six-component force balance. Thrust is the axial force along the z axis of the sensor, and torque is a rotational moment about the z axis. The propeller power coefficient was calculated directly from the measured propeller torque instead of the power supply output to eliminate the losses due to heat generation. The balance bolts to an aluminum extrusion frame secured to the inlet of the plenum. The balance has a maximum of 120 N force on its Z axis and a maximum torque of 2 N/m about its Z axis. The encountered thrust and torque

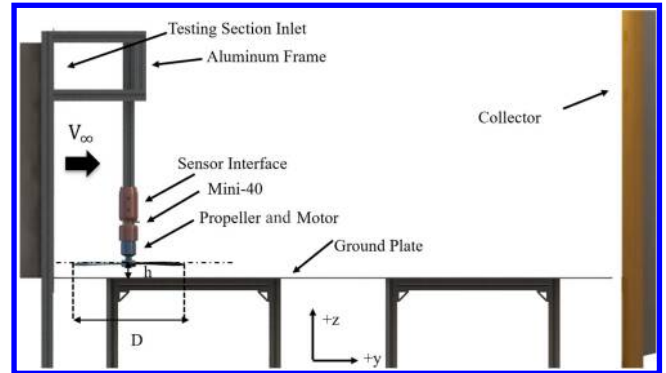


Fig. 1 Schematic of experimental setup of the propeller ground-test stand in the UD-LSWT.

ranges were 5 to 25 N and 0.1 to 0.35 N/m, respectively. The density of the experimental environment is calculated based on the temperature and elevation of the facility using the international standard atmosphere, resulting in a density of 1.189 kg/m^3 .

The data were sampled at the rate of 1000 Hz for a 15 s duration at each data point. Two trials were run in each case to ensure data repeatability. To minimize sensor drift, tare values with the propeller powered off were taken before and after each test. The average of the two tares was subtracted from the respective thrust and torque readings. If the difference between the before and after the tare value was greater than 0.1 N of the force on any axis, the experimental data were then rejected, and the experiments were run again. A finite impulse response band-stop filter [19] was applied to the data to filter out the vibration frequency caused by the propeller rotation and the natural frequency of the aluminum test frame. The tap tests on the test stand were performed with the propeller powered off, which resulted in the structure's natural frequency of 60 Hz. The average uncertainty within the same experimental trail for the force measurement after filtration was $\pm 9.4\%$ across all advanced ratios tested, whereas the average uncertainty within the same experimental trail for the torque measurement was $\pm 7.2\%$. However, the results presented show excellent repeatability from both experimental runs.

Table 1 shows the testing matrix of the experiment. Two off-the-shelf two-blade APC propellers were chosen for the experiment based on their hovering IGE performance in the previous studies of Cai et al. [7,8]. The APC 14×6 propeller represents a low γ/D propeller where the IGE performance benefit is significant in hover, whereas the APC 11×7 propeller represents a medium/high γ/D propeller where the IGE performance benefit is less than the APC 14×6 propeller [7,8]. Three propeller rotational speeds were tested to investigate the Reynolds number effect and the repeatability of the experiment. To achieve different advance ratios, the propeller rotational speed is kept constant, whereas the wind-tunnel freestream velocity is changed accordingly: between 0 and 15 m/s. The propeller ground distance was measured from the center of the propeller hub to the ground plate. This distance was then normalized by the propeller diameter, resulting in the normalized ground distance: h/D . The propeller was mounted at a 90 deg incidence angle θ , where the propeller disk was perpendicular to the freestream, representing

Table 1 Force-based experiment testing matrix

Testing condition	Value	
Propeller	APC 14×6	APC 11×7
Diameter D , m	0.3556	0.2794
Pitch γ , m	0.1524	0.1778
Blade twist angle at 75% radius, deg	10.3	15.1
Number of blades	2	
Solidity σ	0.0808	0.0849
Rotational speed n	83.3, 100, and 116.7 RPS	
Freestream velocity V_∞ , m/s	0–15	
Normalized ground distance (h/D)	0.07–0.86	
Propeller incidence angle θ , deg	75 and 90	

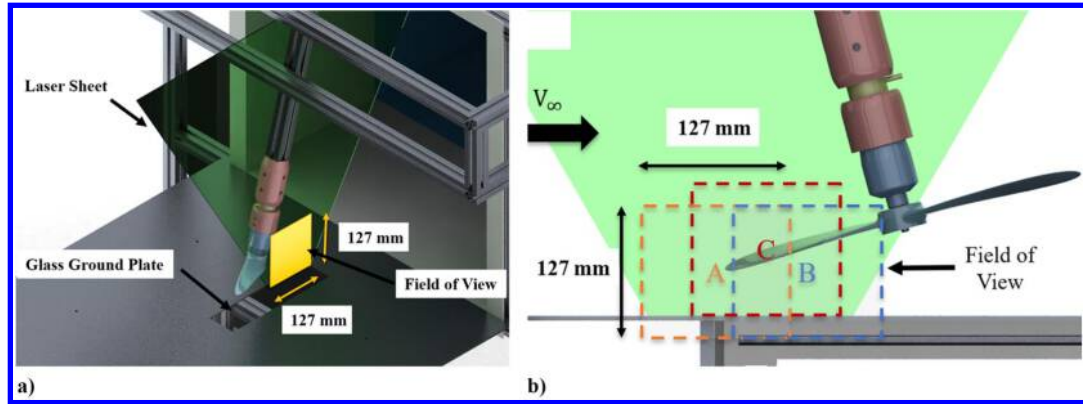


Fig. 2 Schematic of PIV experimental setup in UD-LSWT: a) isometric view, and b) side view.

the low-speed hovering and edgewise flight condition. A 75 deg incidence angle was also tested, which represented a nosedown forward flight condition for the current investigation.

B. Phase-Locked Particle Image Velocimetry

Streamwise phase-locked particle image velocimetry was conducted to further investigate the flowfield between the propeller and the ground at various ground distances and advance ratios. The PIV setup is shown in Fig. 2. A ViCount smoke seeder with glycerin oil was used to generate the seeding for the experiment. A Quantel Twins CFR 300 with a 200 MJ/pulse Neodymium-Doped Yttrium Aluminum Garnet (Nd:YAG) frequency-doubled laser was aligned with a 60 deg fan angle Powell lens, which opened a laser sheet and illuminated the flowfield. A 76.2×304.8 mm glass ground plate was used to let the laser pass through the ground plate. The laser sheet was placed 25 mm away from the centerline of the propeller disk to avoid reflections from the motor and mounting interfaces. An Imperx 2021 charge-coupled device camera was used with a Canon EF 100 mm $f/1.4$ lens, which provided a field of view of 127×127 mm. A 50 mW, 532 nm laser and a photodiode were used to trigger a Quantum Composers (QC) 9600 pulse generator, which was then used to trigger the camera shutter. By changing the delay on the QC pulse generator, the flow passing different propeller phases was captured.

The testing matrix for the PIV experiment is shown in Table 2. One-thousand image pairs were obtained per case and processed using Fluere software. Two iterations were performed in processing with 64-pixel interrogation windows in the first iteration and 32-pixel interrogation windows in the second iteration, with a 50% interrogation window overlap. The time delay between the laser pulses was changed at each advance ratio to obtain 8- to 12-pixel particle displacement at the region of interest in the flowfield. The uncertainty of the PIV measurement, based on the propeller upstream freestream velocity measurement at $J = 0.3$ and 0.4 , was 3.27%. For some of the testing cases, the camera field of view was moved downstream or upward to capture the entire flow feature in-ground effect. This is annotated as fields of view A, B, and C in Fig. 2.

C. Surface Oil Flow Visualization

The oil surface flow visualization was conducted to investigate the flowfield near the ground plate. A mixture of titanium dioxide,

mineral oil, and oleic acid was applied on the ground plate coated with black enamel. The propeller was operating at 90 revolutions per second (RPS), and it was mounted at $\theta = 75$ deg and a ground distance of $0.2h/D$; whereas the freestream velocity varied from 0 to 15 m/s to achieve $0 < J \leq 0.4$. The image was taken immediately after the propeller and wind tunnel were shut down.

III. Results

A. Force-Based Experimental Results

1. Effect of Freestream and Ground Distance on Performance

The results from three different blade Reynolds number tests (83.3, 100, and 116.7 RPS) revealed that the difference in the measured thrust and power coefficients among the cases was less than 5%. Therefore, the experimental result from the 100 RPS case for both propellers was chosen to represent the overall performance changes in the ground effect in forward flight.

Figure 3 shows the normalized thrust coefficient (C_{TIGE}/C_{TOGE}) vs the advance ratio J at different normalized ground distances (h/D) for the APC 14×6 propeller at $\theta = 90$ deg. The C_T of each datapoint is normalized by $C_{TOGE, J=0}$, the C_T at the OGE, and $J = 0$. For the higher h/D cases, an increment in C_T is found at $J > 0.2$ with the increment of J . The thrust augmentation is purely due to the increment in the freestream because the propeller RPS was held constant. The highest h/D case tested, which was $h/D = 0.86$, shows a similar trend as the OGE condition where C_T remains constant within $0 < J < 0.2$ and increases when $J > 0.2$. Similar trends for the fixed-pitch propeller in edgewise flight were also found in a previous study by Cai et al. [20] and the studies conducted by Prudden [21] and Simmons and Hatke [18]. This trend holds until h/D reaches below 0.36, where different trends occur.

At $h/D < 0.29$, the change in C_T vs J does not follow the same trend as seen in the OGE or higher h/D cases. Whereas C_T remains constant at $J < 0.1$, a decrement in C_T is found within the range of $0.2 < J < 0.3$, except for the extreme ground effect cases at $h/D = 0.07$. Eventually, C_T starts to increase after $J > 0.3$ for all cases, indicating a reduction of the ground effect benefit (thrust augmentation), although there is still a 10% increment in thrust when compared to the OGE case. At $J > 0.3$, all cases show a similar trend of increase in C_T , with a clear difference between each h/D case, where a lower h/D results in higher thrust augmentation while at the same time showing a drastic decrement and increment in C_T .

2. Isolating the Effect of Freestream In-Ground Effect

To eliminate the effect of the freestream, an alternative normalization method was used. The C_T of each h/D case was normalized by the C_T at the OGE condition at the same J . As the measured freestream velocity varies slightly between cases, a third-order interpolation of the OGE C_T vs J (with an R^2 value of 0.989) was used in the normalization of each data point in order to increase the accuracy of the data normalization. The same method is also used in all of the remaining data normalizations. The results for the $\theta = 90$ deg case are shown in Fig. 4a. At $h/D \geq 0.86$, the overall trend is similar to

Table 2 PIV testing matrix

Testing condition	Value
Propeller	APC 14×6
Rotational speed n	90 RPS
Fr Freestream velocity V_∞	0–15 m/s
Normalized ground distance (h/D)	0.2 and 0.3
Propeller incidence angle θ	75 and 90 deg
Laser pulse frequency	~15 Hz
Field of view	127×127 mm
Number of image pair	1000/case

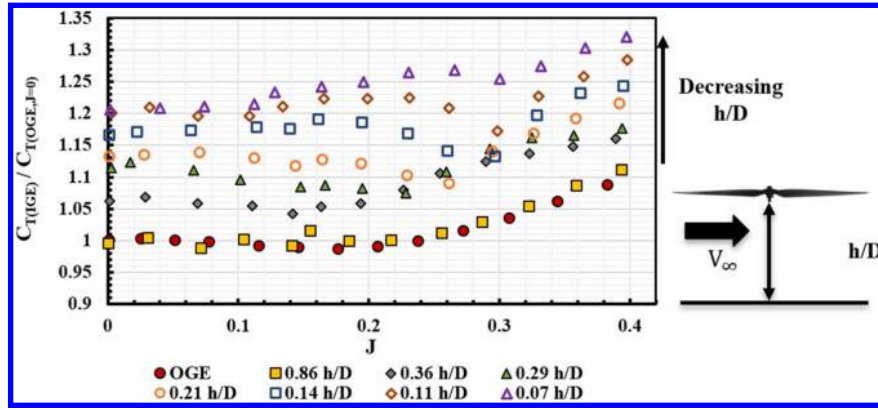


Fig. 3 Normalized thrust coefficient ($C_{TIGE}/C_{TOGE,J=0}$) at $\theta = 90$ deg and 100 RPS in-ground effect for APC 14×6 propeller.

Fig. 3. However, at $h/D \leq 0.14$, a clear increment in C_{TIGE}/C_{TOGE} is observed at $0.1 < J < 0.2$, followed by a clear decrement at $0.2 < J < 0.3$, which was discussed previously. Interestingly, at $J = 0.4$, the value of C_{TIGE}/C_{TOGE} is similar to $J = 0$, indicating that the effect of the freestream in the overall ground effect is negligible beyond a threshold J for each h/D .

Because the data normalization with the OGE at different J values brings more insight into the effect of ground on the propeller performance, the normalization in the following sections will be performed with the same method. An in-depth discussion of the changes in the OGE propeller performance due to the propeller incidence angle was included in a previous study by Cai et al. [20].

Figure 4b shows C_{TIGE}/C_{TOGE} when tilting the propeller forward at $\theta = 75$ deg at different advance ratios. Similar to the overall trend in Fig. 4a, the changes in C_T are minor at $J < 0.1$. A small increment in C_T is again observed within the range of $0.1 < J < 0.15$ at $h/D \leq 0.14$. However, the decrement in C_T occurs at a lower J value when compared to $\theta = 90$ deg; within the range of $0.15 < J < 0.25$. At $J > 0.25$, the overall trend again resembles Fig. 4a. Again, the difference of $C_{TIGE}/C_{TOGE,J}$ between $J = 0$ and $J = 0.4$ is negligible for $h/D > 0.21$, whereas a 4% loss is found for smaller h/D cases.

Similar to C_T , C_P also shows significant departures from the OGE due to the ground effect and advance ratio, as seen in Fig. 5. Again, C_P at each h/D case was normalized by the C_P at the OGE condition

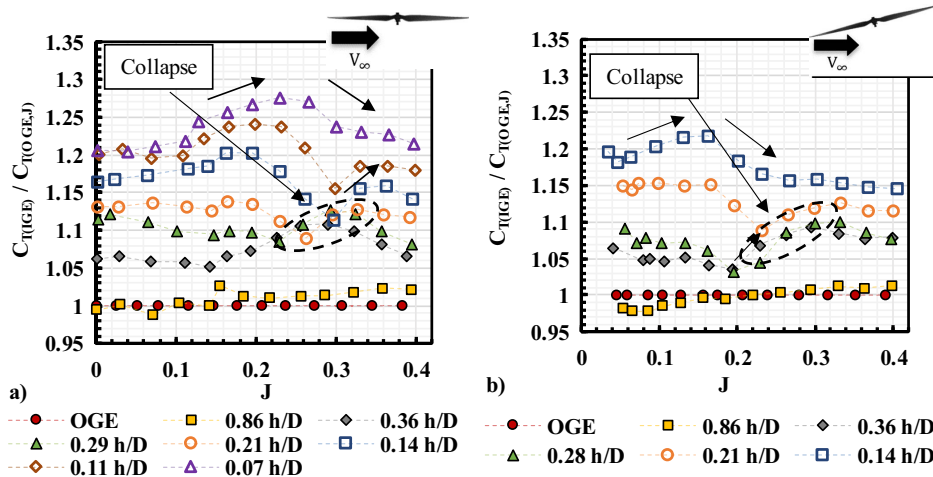


Fig. 4 Normalized thrust coefficient at a) $\theta = 90$ deg and b) $\theta = 75$ deg under 100 RPS in-ground effect for APC 14×6 .

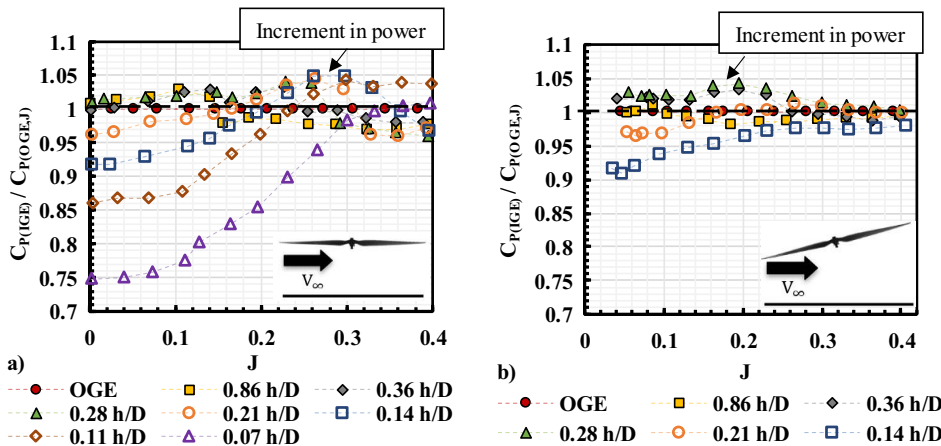


Fig. 5 Normalized power coefficient at a) $\theta = 90$ deg and b) $\theta = 75$ deg under 100 RPS in-ground effect for APC 14×6 .

at the same J , resulting in $C_{P(IGE)}/C_{P(OGE,J)}$. Based on the result from a previous study [7], the ground effect benefit of hovering is noticeable on C_p only at $h/D < 0.28$. This benefit reduces with the increment of J . It is worth noting that for all cases at $h/D < 0.28$, $C_{P(IGE)}/C_{P(OGE)}$ exceeds 1.0 within the range of $0.2 < J < 0.4$, with lower h/D cases extending its maximum $C_{P(IGE)}$ benefit to higher J . At $J \sim 0.4$, the C_p curves for all h/D cases collapse onto each other, indicating a loss of in-ground effect benefits.

Figure 5b shows the normalized power coefficient for $\theta = 75^\circ$. Again, the overall trend resembles Fig. 5a. However, for cases at $h/D < 0.28$, $C_{P(IGE)}/C_{P(OGE)}$ no longer exceeds 1.0 at higher J , whereas the curve for different h/D cases collapses at $J \sim 0.3$, which is lower than $\theta = 90^\circ$. Recall that for C_T at $\theta = 75^\circ$, the changes in C_T also occur at a lower J value, indicating that tilting the propeller forward “advances” the changes in propeller performance with respect to J .

Because the increment in thrust and the decrement in power vary simultaneously in the in-ground and ceiling effects, a common representation of the overall propeller IGE performance in thrust

and power is the “normalized power” (P_{IGE}/P_{OGE}) at constant thrust. The thrust and power measurement data for the APC 14×6 is used as an example to show the calculation process (Fig. 6). First, the propeller IGE and OGE thrust and torque are measured over a range of propeller rotational speeds between 80 and 120 RPS at the same advance ratio. For every h/D of interest, a second-order interpolation for thrust vs RPMs and torque vs RPMs across the rotational speed is created. Then, a thrust level of interest is selected (16 N for this example), and the rotational speed at the selected thrust level is found using the thrust interpolation of thrust vs RPMs at this very h/D . At this particular rotational speed, the IGE power required is obtained from the torque vs RPM interpolation. This process is repeated at every ground distance (h/D) and advance ratio (J) tested.

Multiple thrust levels are selected, and the results are averaged across OGE thrust levels: that is, across propeller rotational speed values. With the freestream velocity known and the propeller rotational speed calculated from the interpolation, the advance ratio is then calculated. The IGE power is then normalized by the power of the OGE at the same thrust level and advance ratio to produce $P_{IGE}/P_{(OGE,J)}$. A thrust level of 16 N is selected in this section for representation, which is the medium thrust level for the APC 14×6 propeller.

The normalized power at constant thrust for $\theta = 90^\circ$ is shown in Fig. 7a. For $h/D \geq 0.86$, the result again resembles the OGE trend. For smaller h/D cases at $J < 0.15$, a slight increment in P_{IGE}/P_{OGE} is observed for all ground distance cases, which is a result of the increment of C_p seen in Fig. 5 at the same J range. With the increment of J , within the range of $0.15 < J < 0.35$, a clear increment of P_{IGE}/P_{OGE} is found for all h/D cases with the increment of J ; then, it collapses with other curves. After the data collapse, a sharp decrease in P_{IGE}/P_{OGE} with the increment of J is observed. An overall difference of 12% in P_{IGE}/P_{OGE} between $J = 0$ and 0.4 is found for extreme ground effect cases at $h/D < 0.14$, due to the loss of ground effect benefit in power reduction, as shown in Fig. 5. However, at higher h/D cases, the difference in P_{IGE}/P_{OGE} between $J = 0$ and 0.4 seems negligible as a result of “thrust recovery” at high J , as seen in Fig. 4.

P_{IGE}/P_{OGE} for $\theta = 75^\circ$ is shown in Fig. 7b, with a very similar trend seen in Fig. 7. As discussed previously, tilting the propeller forward advances the changes in propeller performance with respect to J due to the blade tip being much closer to the ground when compared to the hub. The increment in P_{IGE}/P_{OGE} occurs at $J = 0.1$, whereas the collapse of the curves occurs at $J = 0.25$; both are at lower J when compared to the result for $\theta = 90^\circ$. Again, the difference in P_{IGE}/P_{OGE} between $J = 0$ and 0.4 seems negligible for $h/D > 0.14$, whereas a 10% difference is found at smaller h/D cases.

For a higher γ/D propeller (APC 11×7), the normalized C_T and C_p at $\theta = 90^\circ$ are shown in Fig. 8 at 100 RPS. As discussed in a previous study [7], the ground effect benefit in the thrust and power coefficients reduces when the propeller's γ/D increases; thus, the increment in C_T and decrement in C_p are lower when compared to the

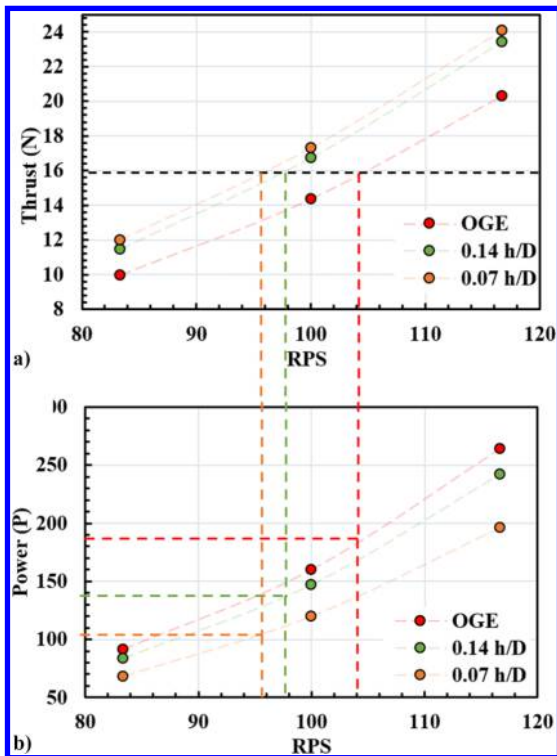


Fig. 6 Representations of a) thrust vs RPS and b) power vs RPS for APC 14×6 propeller at various h/D .

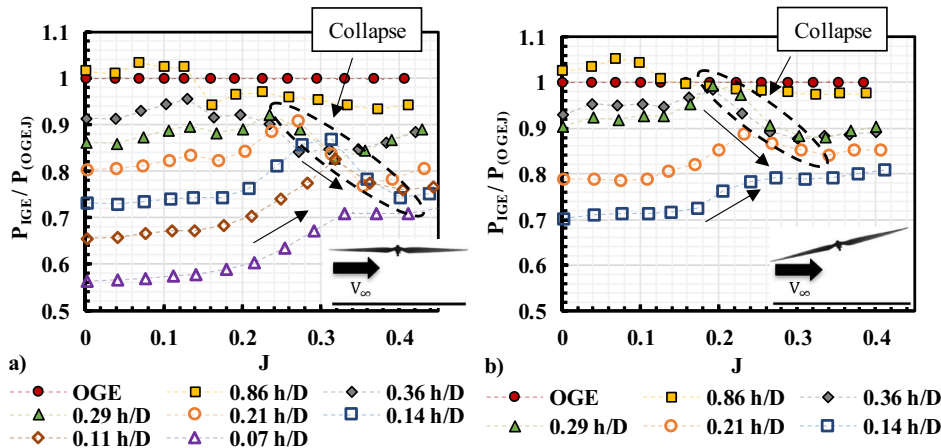


Fig. 7 Normalized power required at constant thrust at a) $\theta = 90^\circ$ and b) $\theta = 75^\circ$ under 100 RPS in-ground effect for APC 14×6 .

APC 14×6 propeller. The overall trend for C_T and C_P for the APC 11×7 propeller is also similar to the APC 14×6 propeller for $h/D > 0.18$. However, a 10% increment in C_P is observed at $h/D \leq 0.18$ at $J > 0.1$, where the C_P curve surpasses the OGE conditions. With a further increment in J , C_P eventually reduces to ~ 1.0 at $J = 0.4$ for $h/D = 0.18$ and $J = 0.6$ for $h/D = 0.09$ based on the results from both 83.3 RPS and 100 RPS testing cases.

To reflect the real flight conditions, the power required at constant thrust is normalized by the OGE condition at hover, resulting in $P_{IGE}/P_{OGE J=0}$, which accounts for the effect of the freestream velocity on the propeller performance. The selected h/D results for both APC 14×6 and APC 11×7 propellers at $\theta = 90$ deg are shown in Fig. 9, along with the data from Cheeseman and Bennett [2] and Curtiss et al. [6].

For the OGE condition for both propellers, with the increment in J , a decrement in $P_{IGE}/P_{OGE J=0}$ is observed as a result of a small increment in C_T at higher J seen in Fig. 3. It is worth noticing that the reduction in $P_{IGE}/P_{OGE J=0}$ at the OGE for the lower γ/D propeller (14×6) is greater than the high γ/D propeller (11×7). This indicates a higher performance for the lower γ/D propeller in the edgewise flight. The lower ground effect benefits in terms of the power decrement for the 14×6 propeller are due to the occurrence of blade stall at the high-speed edgewise flight for the high γ/D fixed-pitch propeller. A similar trend is also observed in a previous study [20]. For both propellers' in-ground effects, the overall trend in the

change of $P_{IGE}/P_{OGE J=0}$ is similar to Fig. 7. Again, with a higher γ/D propeller, the ground effect benefit reduces in hover and forward flight conditions when compared to a lower γ/D propeller as a result of lower IGE thrust augmentation and power savings.

When compared to the data from Cheeseman and Bennett [2], the reduction in $P_{IGE}/P_{OGE J=0}$ for the OGE condition with the increment in J is much greater than the OGE condition for both propellers tested. This is likely a result of the variable/cyclic blade pitch mechanism of the helicopter rotor, which reduces the drag on the advancing blade and the overall power consumption of the rotor. Moreover, the rotors in the literature were tested in a much higher Reynolds number than the current study, which also contributes to the better performance observed. It is also worth noting that the propeller chosen in the current study is not optimized for edgewise flight. When compared to the conventional helicopter blades, the blade twist angle for both the propellers tested is significantly higher, which also contributes to higher power consumption in the edgewise flight. For both literatures [2,6], the reduction in $P_{IGE}/P_{OGE J=0}$ in the IGE occurs at $J < 0.15$, which is lower than both propellers tested (at $J = \sim 0.25$). With a further increment in J , the difference in $P_{IGE}/P_{OGE J=0}$ at different h/D conditions reduces, which agrees with the experimental result for the propeller tested.

B. Ground Effect Phase-Lock PIV

A phase-locked PIV experiment was conducted to investigate the flowfield around the APC 14×6 propeller in-ground effect. The

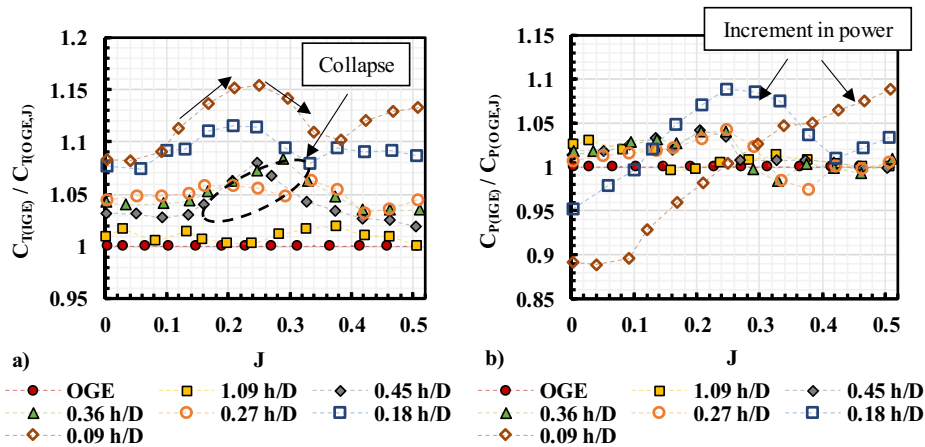


Fig. 8 Representations of a) normalized thrust coefficient and b) normalized power coefficient at $\theta = 90$ deg and 100 RPS for APC 11×7 propeller.

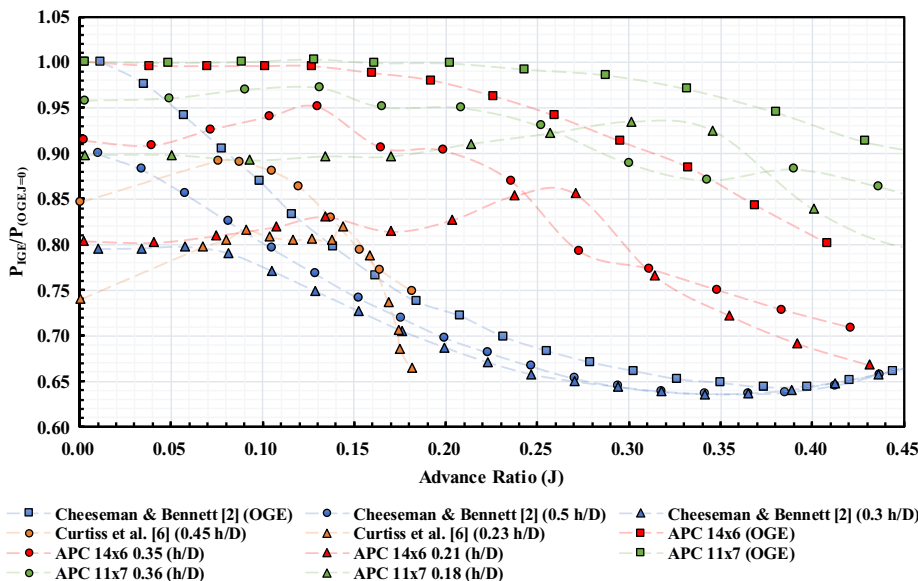


Fig. 9 OGE hover normalized power required at constant thrust for APC 14×6 (15 N) and APC 11×7 (8 N) propellers at $\theta = 90$ deg.

images are taken at the same propeller phase, where the propeller blade rotates approximately 80 deg away from the laser sheet. All contours presented in this section are averaged over several instantaneous images acquired when the propeller blade was at the exact same location every six rotations. The vortex centers of the propeller tip vortex and the ground vortex were tracked for each image based on the maximum Q criterion. If the location of the vortex center was outside the 95% confidence interval on either dimension, the image was removed from the analysis. An average of 4.2% of the images were eliminated during this process. The measured vorticity in the PIV image was normalized by the propeller diameter and the propeller blade speed at the 75% radius location, which is written as

$$\omega_{\text{norm}} = \frac{\omega * D}{n * \pi D * 0.75} \quad (1)$$

Figures 10a and 10b show the flow visualization at $J = 0$ and $J = 0.1$ for incidence angles of $\theta = 90$ deg and $\theta = 75$ deg at $h/D = 0.2$. At $J = 0$, the tip vortices spread radially in all directions due to the ground effect for both θ cases. With the increment in J , for the $\theta = 75$ deg case, the momentum of the freestream increases. The radially extending propeller wake is then pushed back by the free-stream and enveloped in itself, creating a fountain vortex at $J = 0.2$. This fountain vortex also forms for the $\theta = 90$ deg case, although the overall size of the fountain vortex is larger than the one for the $\theta = 75$ deg case because the camera field of view cannot capture the entire fountain vortex. The evidence of the fountain vortex is also captured by the surface flow visualization discussed in the next section.

Recall from the C_T result in Fig. 4 that a decrement in C_T is found within the range of $0.2 < J < 0.3$ for $\theta = 90$ deg and $0.15 < J < 0.25$ for $\theta = 75$ deg, which correlate to the formation of this fountain

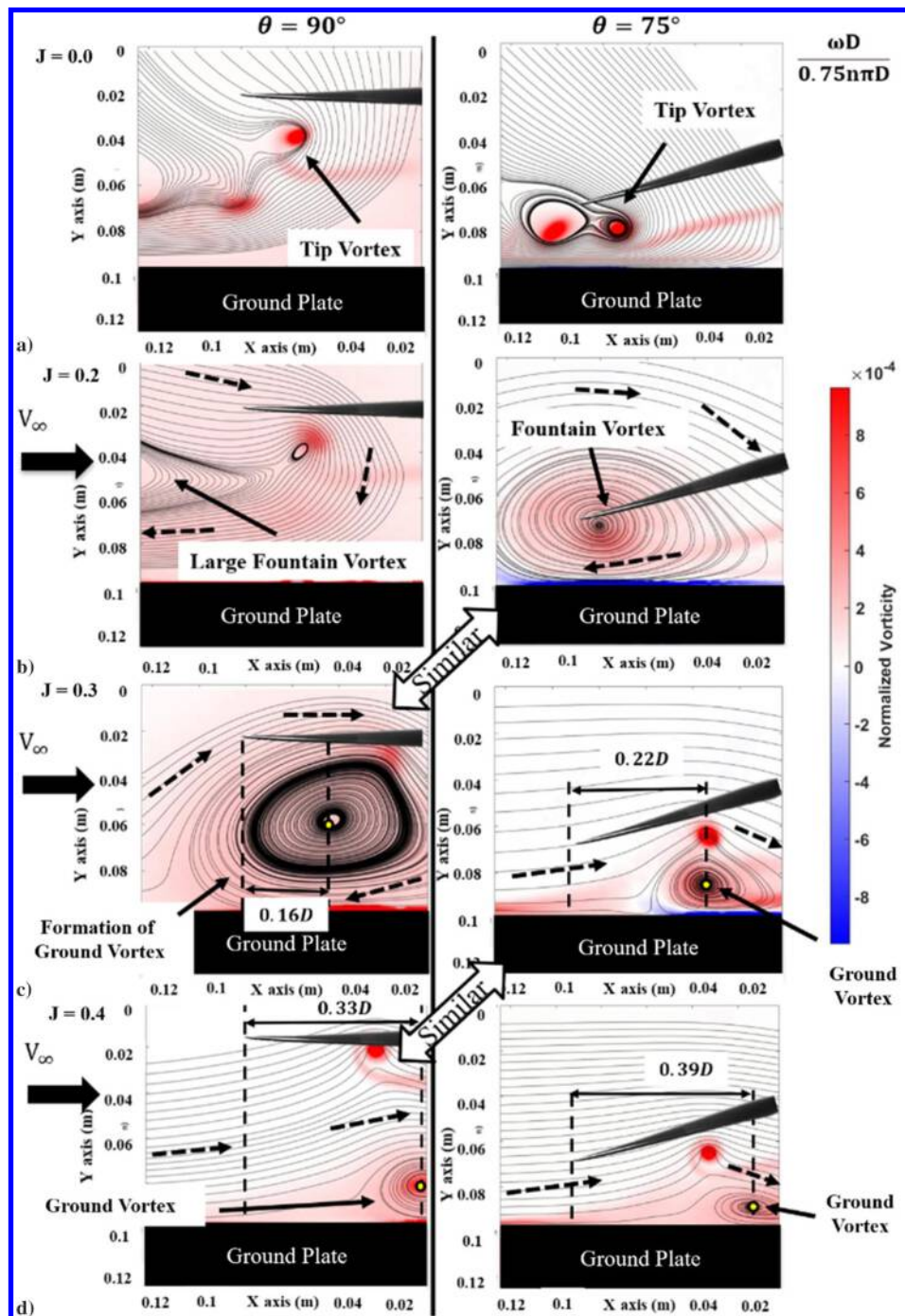


Fig. 10 Flow visualization for $\theta = 90$ deg (left) and $\theta = 75$ deg (right) at a) $J = 0$ and b) $J = 0.2$ and $h/D = 0.2$.

vortex. The fountain is the recirculation region formed by the propeller wake being injected back into the propeller disk [6,11,22], which results in a loss of thrust generation, as seen in Fig. 4.

Upon further increasing J , as seen in Figs. 10c and 10d, the size of the fountain vortex reduces; and a stationary ground vortex is formed between the propeller and the ground for both the $\theta = 75$ deg case and the $\theta = 90$ deg case at $J > 0.3$. Recall that the C_T and C_P differences between each h/D case are relatively small at this advance ratio range (Figs. 4 and 5), indicating a reduction in the ground effect on propeller performance. For both incidence angle cases, the location of the ground vortex moved further downstream at $J = 0.4$ when compared to $J = 0.3$. Moreover, the location of the ground vortex also moved further downstream for the $\theta = 75$ deg case when compared to the $\theta = 90$ deg case: from $0.16D$ away from the propeller tip to $0.22D$ at $J = 0.3$, and from $0.33D$ to $0.39D$ at $J = 0.4$. It is also worth noting that the flowfield for the $\theta = 90$ deg case at $J = 0.3$ is similar to the flowfield for the $\theta = 75$ deg case at $J = 0.2$, whereas the flowfield for the $\theta = 90$ deg case at $J = 0.4$ is similar to the flowfield for the $\theta = 75$ deg case at $J = 0.3$. This indicates that reducing θ in the in-ground effect under certain J would result in a similar flowfield at a higher J for a higher θ case.

Figure 11 emphasizes the effect of the ground distance on the propeller flowfield in the ground effect. At $J = 0.3$, the ground vortex occurs at $0.16D$ away from the propeller tip for (h/D) = 0.2; whereas the location of the vortex moves downstream at $h/D = 0.3$, where the distance between the ground vortex to the propeller tip increases to $0.19D$. The same phenomena are also observed at $J = 0.4$, where the location of the ground vortex for $h/D = 0.3$ moves to $0.4D$ from the propeller tip (where the camera location has to change to capture the ground vortex); the strength of the ground vortex also reduces significantly when compared to $h/D = 0.2$, based on the calculated vorticity.

Compared to the results in Figs. 10c and 10d, the changes in the in-ground vortex location due to the increment of h/D are similar to the effect of the increment of J when h/D is held constant, where a similar conclusion is also found in the study conducted by Curtiss et al. [6]. Moreover, as mentioned previously, the location of the ground vortex also moves downstream when the propeller incidence angle reduces. In other words, the effect of increasing h/D has dynamic similarities to the effect of decreasing θ or increasing J in the in-ground effect.

C. Surface Oil Flow Visualization

Surface oil flow visualization is conducted to further visualize the presence of the fountain vortex and ground vortex. The oil flow visualization results for $\theta = 75$ deg and $h/D = 0.2$ at various J are shown in Fig. 12. At a small J value, as the propeller wake dominates, the oil pattern shows an outward radial flow from the propeller. With the increment in J , a stagnation streamline is formed upstream of the propeller disk as a result of the fountain vortex, seen in Fig. 10b. The fountain vortex transformed into a steady ground vortex with a further increment of J , as seen in Fig. 10c. The ground vortex stagnates the oil around the propeller and resembles the shape of a horseshoe vortex. The location of this horseshoe vortex moves downstream, and its width reduces simultaneously with the increment in J .

D. Overall Effect of Flowfield on Propeller Performance In-Ground Effect

Figure 13 separates the propeller performance in-ground effect in forward flight into three different regions based on the force-based experiment and the phase-locked PIV experiment result discussed previously. At a small J range, where the propeller wake dominates, the ground effect benefit is at its optimal state, resembling that of the hover condition. A further increment in J initiates the formation of the fountain vortex, which disrupts the thrust produced and increases the power required. The threshold of the fountain vortex formation is set at the point where P_{IGE}/P_{OGE} increases when compared to the hovering condition; the advance ratio where this threshold occurs increases with the decrement in h/D , as shown in Fig. 13. This also agrees with the PIV result discussed in Sec. III.B. The transition between the fountain vortex and ground vortex occurs right after the peak of P_{IGE}/P_{OGE} , where the strength of the fountain/ground vortex reaches its peak [10], resulting in a significant reduction of the propeller performance in-ground effect. With a further increment of J , a ground vortex formed in between the propeller disk and the ground where the ground effect benefit is reduced when compared to the hovering condition.

Figure 14 summarizes the different flow regions shown in Fig. 13 for both $\theta = 90$ deg and $\theta = 75$ deg, in the format of J vs h/D , based on the threshold discussed in Fig. 13. At $h/D > 0.5$, the IGE benefit is negligible. For $h/D < 0.5$, at a higher h/D , the onset of fountain vortex formation and the ground vortex formation occurs at a

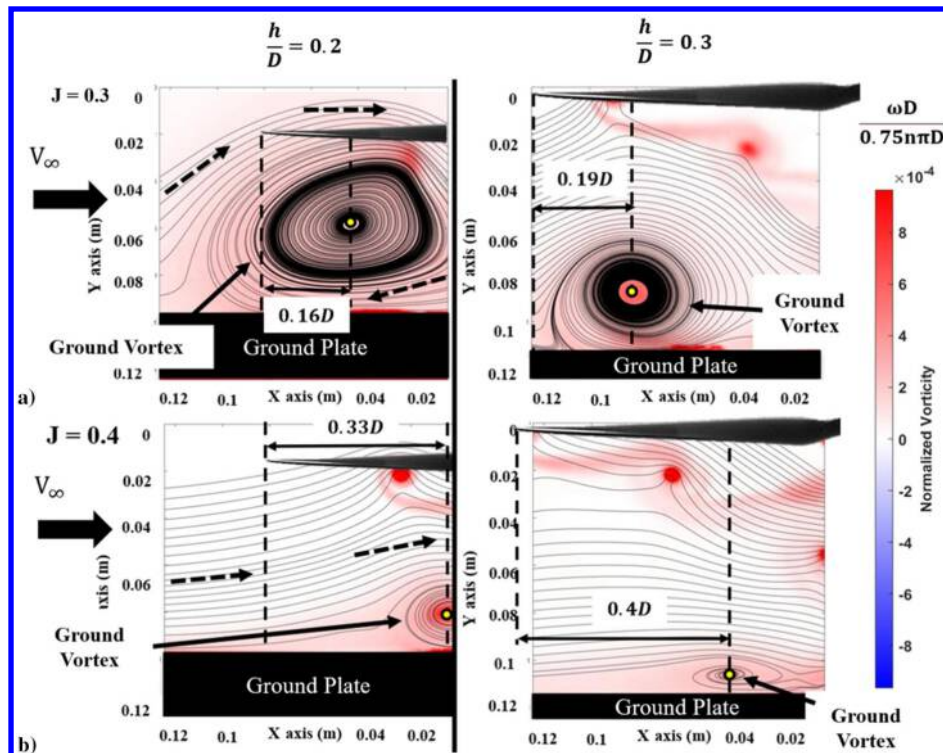


Fig. 11 Flow visualization for $\theta = 90$ deg at $J = 0.3$ (left) and $J = 0.4$ (right) at a) $h/D = 0.2$ and b) $h/D = 0.3$.

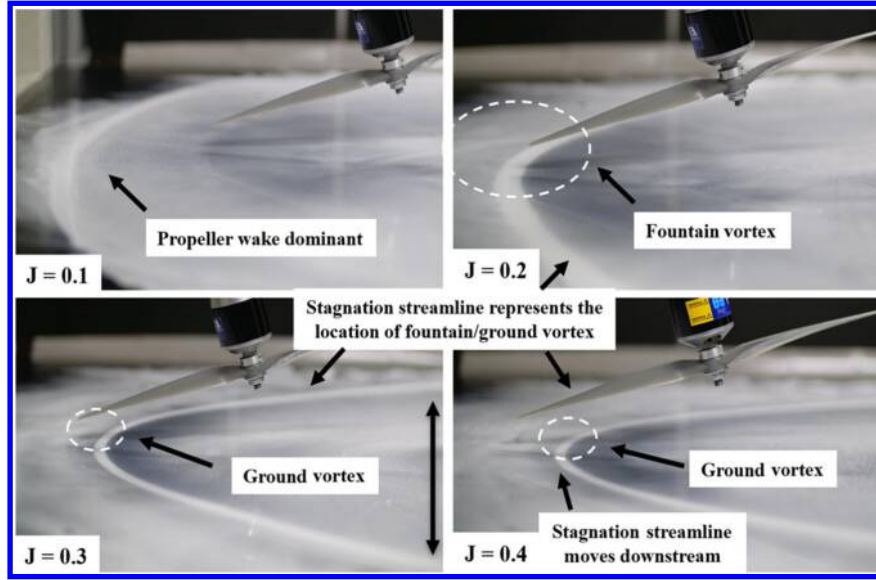


Fig. 12 Surface oil visualization for $\theta = 75^\circ$ and $h/D = 0.2$ within the range of $0.1 \leq J \leq 0.4$.

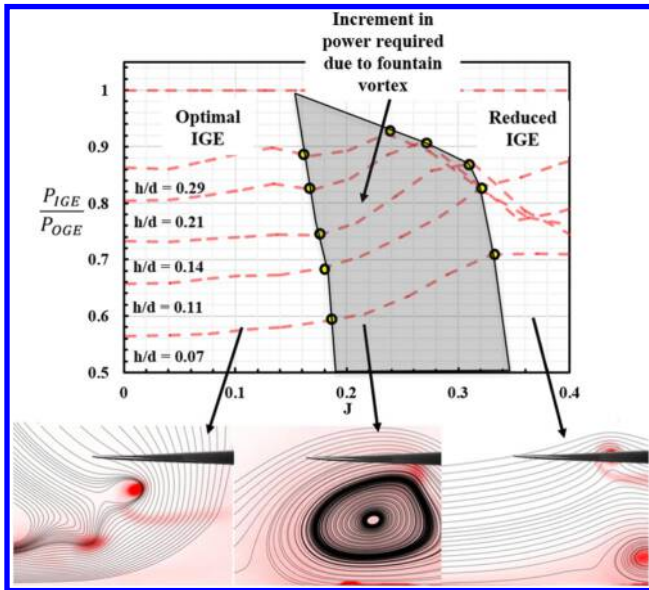


Fig. 13 Zones of propeller performance in-ground effect forward flight marked by three distinct flowfields.

lower advance ratio. Decreasing the h/D delays the onset of these vortices to a comparatively higher advance ratio. When the incidence angle is reduced, the onset of the fountain vortex and the ground vortex occurs at significantly lower advance ratios as compared to $\theta = 90^\circ$ deg, which is also shown in the PIV results in Fig. 10.

A similar plot is also discussed in Curtiss et al.'s work [6], which used the normalized advance ratio J^* defined by Eq. (2). As mentioned previously, J^* also represents the ratio of the freestream velocity V_∞ and the propeller-induced velocity V_i :

$$J^* = J \sqrt{\frac{\pi}{2C_T}} = \frac{V_\infty}{nD} \sqrt{\frac{\pi}{2(V_i^2 \frac{\pi}{2n^2 D^2})}} = \frac{V_\infty}{V_i} \quad (2)$$

From Eq. (2), J^* represents the effective propeller wake skew angle χ_{eff} at the OGE as shown in Fig. 15. To calculate χ_{eff} , the classic momentum theory can be used when assuming the propeller wake velocity equals two times the inflow velocity [16]. Hence, $V_{w_y} = 2V_i$ and $V_{w_x} = 2V_\infty$. With the schematic shown in Fig. 15, χ_{effOGE} can be estimated from the contribution of the propeller wake skew angle χ_{prop} and the propeller forward tilt angle δ as

$$\chi_{\text{effOGE}} \cong \chi_{\text{prop}} + \delta \quad (3)$$

where

$$\chi_{\text{prop}} = \text{atan}\left(\frac{V_{w_x}}{V_{w_y}}\right) = \text{atan}\left(\frac{2(V_\infty \sin \theta)}{2(V_i + V_\infty \cos \theta)}\right) \quad (4)$$

and

$$\delta = 90 - \theta \quad (5)$$

At $\theta = 90^\circ$ deg and $\delta = 0^\circ$ deg, Eq. (3) can be simplified to

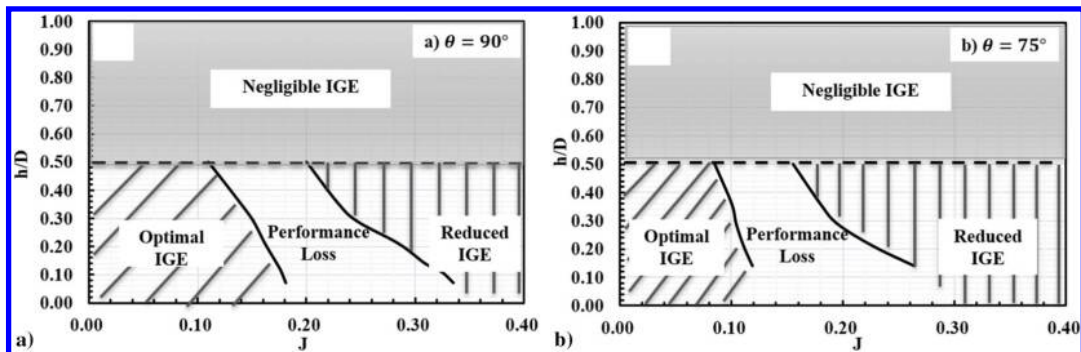


Fig. 14 Propeller IGE performance region for a) $\theta = 90^\circ$ deg and b) $\theta = 75^\circ$ deg, plotting with J vs h/D .

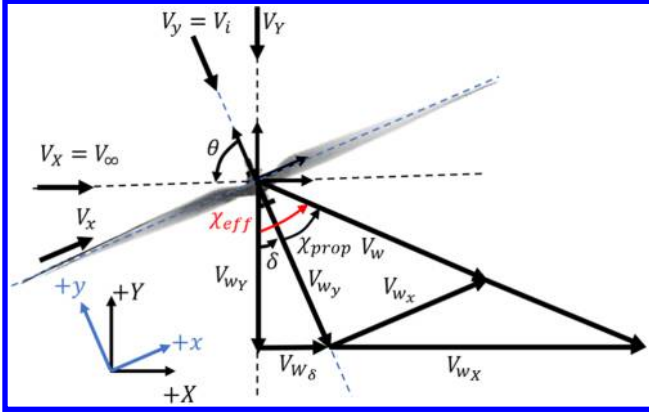


Fig. 15 Schematic showing estimation of OGE effective wake angle.

$$\chi_{\text{eff OGE}}|_{\theta=90 \text{ deg}} = \text{atan}\left(\frac{V_{\infty}}{V_i}\right) = \text{atan}(J^*) \quad (6)$$

Hence, for a pure edgewise flight, $J^* = 1$ indicates the wake velocity matches that of the induced velocity, resulting in $\chi_{\text{eff OGE}} = 45 \text{ deg}$. $J^* > 1$ means that $\chi_{\text{eff OGE}}$ is greater than 45 deg and the freestream momentum becomes dominant. At $J^* < 1$, the propeller wake is dominant when compared to the freestream. Rewriting in the inertial frame of reference (X, Y) , Eq. (3) can be written as

$$\chi_{\text{eff OGE}} \cong \chi_{\text{prop}} + \delta = \text{atan}\left(\frac{V_{\infty} + V_{w\delta}}{V_{wy}}\right) \quad (7)$$

Using Fig. 16,

$$\chi_{\text{eff OGE}} = \text{atan}\left(\frac{2V_{\infty} + 2V_i \cos \theta}{2V_i \sin \theta}\right) \quad (8)$$

In terms of J^* ,

$$\chi_{\text{eff OGE}} = \text{atan}\left(\frac{J^*}{\sin \theta} + \tan(\delta)\right) \quad (9)$$

Equation (9) shows the sensitivity of the wake skew angle to J^* and θ . A similar $\chi_{\text{eff OGE}}$ can be achieved at either a higher J^* with a low θ or at a lower J^* with a higher θ . Moreover, a similar $\chi_{\text{eff OGE}}$ will also result in a similar OGE and IGE wake flowfield, which would affect the overall propeller performance in-ground effect.

The results in Fig. 14 are replotted with respect to J^* in Fig. 16, along with the boundaries of the flow regions annotated. The bound-

daries of the flow regions introduced by Curtiss et al. [6] for a conventional rotor are shown in black shaded area and dashed lines between $0.4 < J^* < 1.0$. As mentioned previously, at $J^* < 0.5$, the propeller flow is dominant; and the flow pattern is similar to the hovering condition. Within the range of $0.5 < J^* < 0.7$, the formation of the fountain vortex is observed and a loss in the rotor ground effect performance is measured; a further increase in J^* will result in the formation of a ground vortex. The upper boundary of J^* for the ground vortex occurs at $J^* \sim 1.0$, where at higher J^* , there is no flow pattern observed [6,23]

The boundary of the flow pattern for the current study (Fig. 14) is shown in Fig. 16 with orange shaded area representing $\theta = 75 \text{ deg}$ and blue shaded area representing $\theta = 90 \text{ deg}$. The overall trend for the boundary with respect to J^* and h/D is similar to the result from Curtiss et al. [6]; that is, with the reduction of h/D , the boundary moves to a high J^* . However, the J^* range where the fountain vortex and ground vortex are observed is wider than the region found in the study by Curtiss et al. [6]. However, the current study did not quantify the upper bound of J^* where the ground vortex vanishes.

It is obvious that similar flowfield regions do not collapse within the range of the $\theta = 75$ and 90 deg cases. Based on Eq. (9), a higher θ will result in a lower $\chi_{\text{eff OGE}}$ that is the same J^* . And, a similar $\chi_{\text{eff OGE}}$ will result in a similar wake flowfield. To achieve the same $\chi_{\text{eff OGE}}$ for both $\theta = 75$ and 90 deg cases, the latter one needs to operate at a higher J^* ; this agrees with the trend seen in Fig. 16 where a similar flowfield for $\theta = 90 \text{ deg}$ occurs at a higher J^* . This also agrees with the PIV results discussed in Fig. 10.

It is highly likely that the propeller IGE flowfield has a strong connection with the propeller's overall IGE performance. Hence, the IGE propeller performance trend at different θ in edgewise flight can be estimated using $\chi_{\text{eff OGE}}$ by calculating the $\theta = 90 \text{ deg}$ equivalent J_s^* or, in other words, a scaled normalized advance ratio J_s^* . Based on Eq. (9) and Fig. 15, J_s^* can be calculated when considering

$$\chi_{\text{eff OGE}}|_{\theta=90 \text{ deg}} = \chi_{\text{eff OGE}} \theta$$

where

$$\text{atan}\left(\frac{J_s^*}{\sin 90 \text{ deg}} + \tan 0 \text{ deg}\right) = \text{atan}\left(\frac{J^*}{\sin \theta} + \tan \delta\right) \quad (10)$$

Solving for J_s^* yields

$$J_s^* = \frac{J^*}{\sin \theta} + \tan \delta \quad (11)$$

The experimental results in Fig. 16 are represented in terms of J_s^* in Fig. 17. Although the flowfield boundary for the $\theta = 75 \text{ deg}$ case in terms of J_s^* gets closer to the $\theta = 75 \text{ deg}$ case, both boundaries do

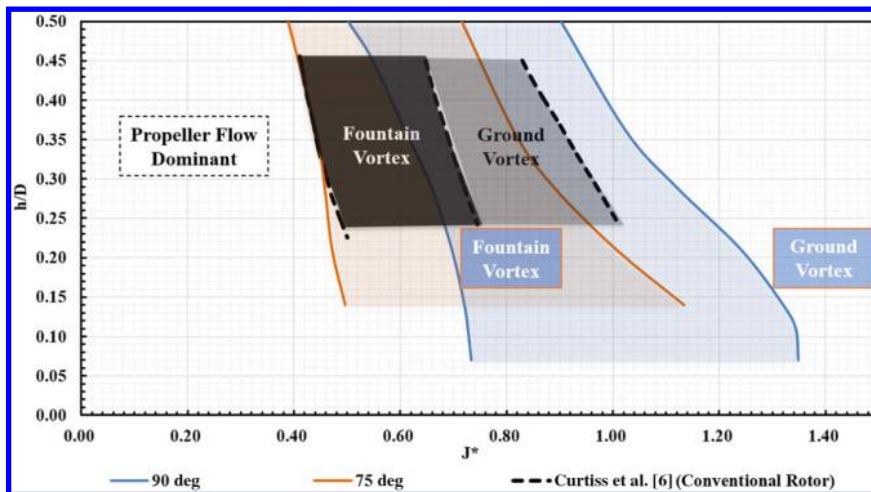


Fig. 16 Propeller IGE flow pattern region plotted with normalized advance ratio and the result from Curtiss et al. [6].

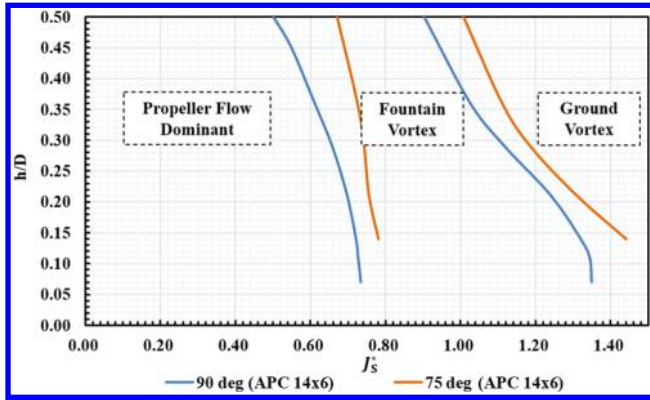


Fig. 17 Propeller IGE flow pattern region plotted with equivalent advance ratio J_s^* .

not overlap on each other, indicating the estimation based on Eqs. (9) and (11) is not adequate. Hence, a correction method for the IGE wake skew angle $\chi_{\text{eff IGE}}$ is required.

Recall from the PIV results in Figs. 10 and 11, as well as the surface oil visualization result in Fig. 12, that a stagnation point was observed on the ground plate when the fountain vortex and the ground vortex are formed. It is hypothesized that this results in a higher effective propeller IGE wake skew angle $\chi_{\text{eff IGE}}$, in comparison to $\chi_{\text{eff OGE}}$ discussed previously. The momentum of the incoming freestream between the propeller and the ground plate is acting on the propeller wake, which further pushes the propeller wake downstream when compared to the OGE condition. This results in an additional velocity component in the IGE wake, as indicated in Fig. 18. The addition of V_∞ in the wake increases χ_{eff} at a given J^* when compared to the results shown in Fig. 15. Hence, $\chi_{\text{prop IGE}}$ consists of the contribution from the propeller incidence angle $V_{W\delta}$, the freestream of propeller inflow V_{wX} , and the freestream between the propeller wake and the ground plate V_∞ .

Using a similar approach as shown in Eq. (5), the IGE propeller effective wake angle $\chi_{\text{eff IGE}}$ can be estimated as

$$\chi_{\text{eff IGE}} \cong \chi_{\text{prop IGE}} + \delta = \text{atan}\left(\frac{2V_\infty + V_\infty + 2V_i \cos \theta}{2V_i \sin \theta}\right) \quad (12)$$

In terms of J^* ,

$$\chi_{\text{eff IGE}} = \text{atan}\left(\frac{3}{2} \frac{J^*}{\sin \theta} + \tan \delta\right) \quad (13)$$

The differences between the OGE and IGE wake skew angles predicted by Eqs. (9) and (13), respectively, are shown in Fig. 19. It can be seen in Fig. 19 that at a given inclination angle, a similar OGE wake skew angle is achieved at a lower J^* in in-ground-effect conditions. The results from Fig. 19 agree well with the trends shown in the experimental results seen in Fig. 16, where a similar flowfield occurs at a lower J^* at a lower θ .

Therefore, J_s^* in Eq. (11) can be rewritten to account for $\chi_{\text{eff IGE}}$ as J_{SIGE}^* . Similar to the previous derivation, using Eq. (13) and Fig. 18, J_{SIGE}^* can be calculated when considering

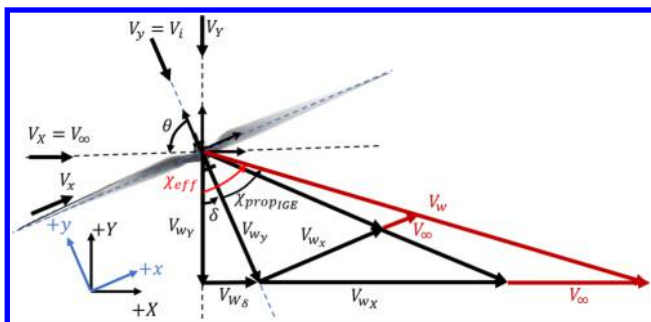


Fig. 18 Schematic showing estimation of IGE effective wake angle.

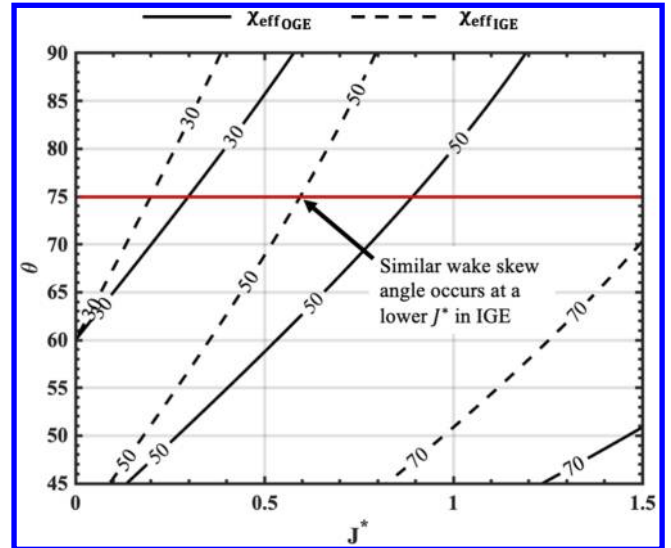


Fig. 19 Differences between OGE and IGE wake skew angles as functions of J^* and θ .

$$\chi_{\text{eff IGE}}|_{\theta=90 \text{ deg}} = \chi_{\text{eff IGE}}|_{\theta}$$

where

$$\text{atan}\left(\frac{3}{2} \frac{J_{\text{SIGE}}^*}{\sin 90 \text{ deg}} + \tan 0 \text{ deg}\right) = \text{atan}\left(\frac{3}{2} \frac{J^*}{\sin \theta} + \tan \delta\right) \quad (14)$$

Solving for J_{SIGE}^* yields

$$J_{\text{SIGE}}^* = \frac{J^*}{\sin \theta} + \frac{2}{3} \tan \delta \quad (15)$$

The contour of the wake skew angle J_{SIGE}^* vs θ from Eq. (15) is shown in Fig. 20. It is evident that the sensitivity of J_{SIGE}^* and θ was normalized in such a way that the same wake skew angle is obtained at a given J_{SIGE}^* . The experimental results in Fig. 16 can now be represented in terms of J_{SIGE}^* and shown in Fig. 21. It can be seen that the different θ collapses. Therefore, irrespective of the propeller incidence angles, similar wake patterns can be observed for the same J_{SIGE}^* . In this case, the overall propeller IGE performance trend under a different edgewise θ range can be estimated by using the propeller performance at one known θ case.

$P_{\text{IGE}}/P_{\text{OGE},J}$ for the APC 14 $\times$ 6 propeller at $\theta = 90$ and 75 deg (shown in Fig. 7) is replotted in terms of J_{SIGE}^* in Fig. 22 for the

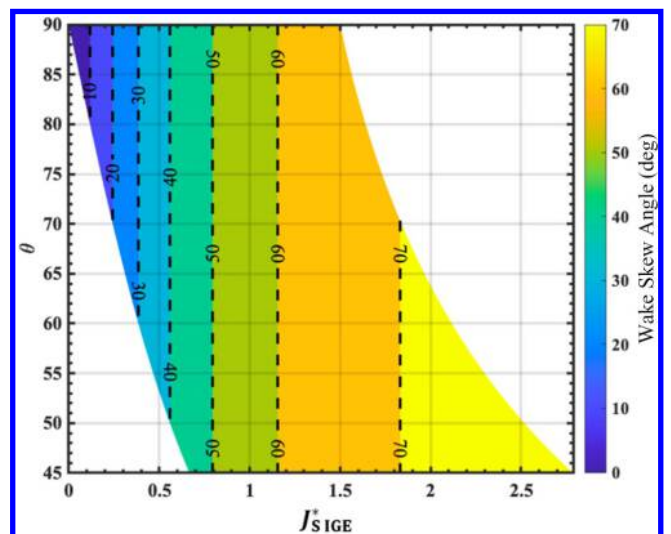


Fig. 20 IGE wake skew angle as a function of θ and J_{SIGE}^* .

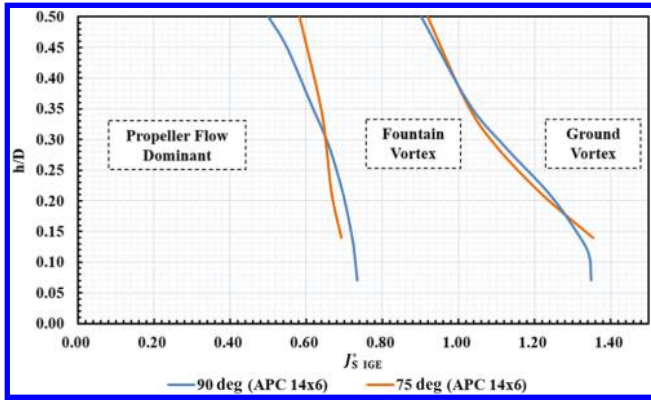


Fig. 21 Propeller IGE flow pattern region plotted with equivalent advance ratio $J_{S IGE}^*$.

selected case at $h/D < 0.5$. Despite the differences seen in the force-based experiment and the PIV images, the overall trend in $P_{IGE}/P_{OGE,J}$ for both θ cases at the same h/D occurs at similar $J_{S IGE}^*$. At $J_{S IGE}^* < \sim 0.7$, the propeller flow dominates, where the value of $P_{IGE}/P_{OGE,J}$ remains relatively constant. Within the range of $\sim 0.7 < J_{S IGE}^* < \sim 1.2$, the fountain vortex occurs along with an increment in $P_{IGE}/P_{OGE,J}$. At $J_{S IGE}^* > \sim 1.2$, the ground vortex is observed where the value of $P_{IGE}/P_{OGE,J}$ decreases with the increment in $J_{S IGE}^*$. Hence, with the propeller performance at one known θ , which can be modeled by different methods [2,5,6] or measured in the experiment, the propeller performance trend at the θ of interest vs J^* can be estimated by Eq. (15). The equivalent flight advance ratio J for this specific θ condition can then be backcalculated by Eq. (2). Therefore, the range of J where the fountain vortex and ground vortex occur at different θ , as well as the corresponding changes in the propeller performance, can be estimated. Therefore, the propeller-driven UAV operation can be designed to avoid the forward flight speed range where significant performance loss occurs at certain forward flight θ conditions.

IV. Conclusions

The changes in propeller performance at different incidence angles in hover and forward flight were studied in the ground effect. The effect of J on the propeller performance was negligible at $J < 0.1$ for all h/D , where the propeller wake is dominant. With the increment in J , at $h/D < 0.3$, a fountain vortex was formed, resulting in a reduction of the propeller thrust generated and an increment in the power required, and thus a reduction in propeller performance. A further increment of the advance ratio resulted in the formation of a steady ground vortex between the propeller disk and the ground plate, where the ground effect benefits in both thrust and power were minimized.

The location of the ground vortex moved downstream with the increment of J , as evidenced by phase-locked PIV and surface oil flow visualization. At higher h/D , this ground vortex vanished, resulting in a performance similar to that of the OGE condition.

Similar to the ground effect in hovering, a lower γ/D propeller has a higher IGE benefit in forward flight. Even though the overall performance change of the propeller at each flow regime is similar to the general trend of the helicopter rotors, the formation of the fountain and ground vortices occurs at a higher J^* for a propeller when compared to a conventional rotor.

Increasing the propeller h/D or tilting the propeller forward (reducing θ) would result in the aforementioned flow pattern occurring at a lower J ; the latter was a result of increasing $\chi_{eff,IGE}$ of the rotor. At the same $\chi_{eff,IGE}$, the IGE flow pattern at different θ was similar. A scaled J^* for the IGE condition $J_{S IGE}^*$ was proposed, which delivered the same wake skew angle independent of the incidence angle θ and provided an estimation of the propeller IGE performance at different θ .

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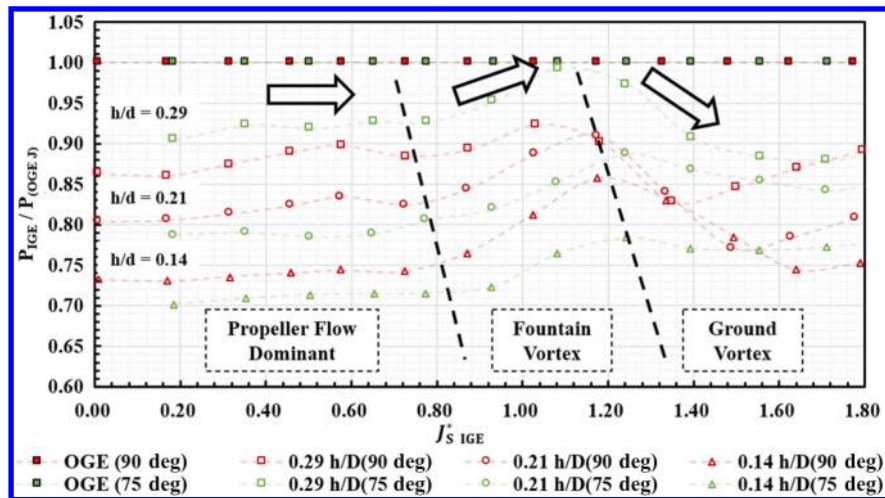


Fig. 22 Normalized power required at constant thrust vs $J_{S IGE}^*$ at $\theta = 90$ deg and $\theta = 75$ deg for APC 14×6 propeller.

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